

1

a)

	Red	blue	Green	grey
Red	1	0	0	0
blue	0	1	0	0
blue	0	1	0	0
red	1	0	0	0
green	0	0	1	0
grey	0	0	0	1

$$X_{\text{Onehot}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

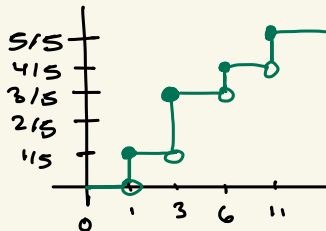
b) 6 Samples

$$u = 2/6 \quad u = 2/6 \quad u = 1/6 \quad u = 1/6 \quad \text{total} = 1$$

Prob: $\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{6}$ $\frac{1}{6}$

c) hist: [1, 3, 3, 6, 11] → Already sorted

$$\hat{F}(x) = \begin{cases} 0 & x \leq 1 \\ 1/5 & 1 < x < 3 \\ 3/5 & 3 \leq x < 6 \\ 4/5 & 6 \leq x < 11 \\ 1 & x \geq 11 \end{cases}$$



$$\begin{aligned} \hat{F}(1) &= 1/5 \\ \hat{F}(3) &= 3/5 \\ \hat{F}(6) &= 4/5 \\ \hat{F}(11) &= 1 \end{aligned}$$

2.

$$y = a + bx$$

a)

$$F_Y(y) = P(Y \leq y)$$

$$P(a + bx \leq y)$$

$$x \leq \frac{y-a}{b}$$

$$P(Y \leq y) = P\left(x \leq \frac{y-a}{b}\right) \Rightarrow F(x)\left(\frac{y-a}{b}\right)$$

der.

$$\frac{d}{dy} F(x)\left(\frac{y-a}{b}\right) \Rightarrow f(x)\left(\frac{y-a}{b}\right) \frac{1}{b} \Rightarrow \frac{1}{b} f(x) \frac{y-a}{b}$$

b)

$$\frac{1+3+4}{3} = \frac{8}{3} = 2.67$$

c)

$$1^2 = 1 \quad 3^2 = 9 \quad 4^2 = 16$$

$$\frac{1+9+16}{3} = \frac{26}{3} = 8.67$$

d)

$$\left(\frac{8}{3}\right)^2 = \frac{64}{9} = 7.11$$

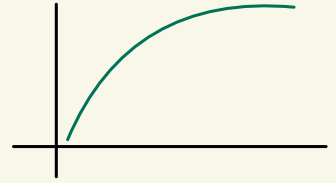
$$\text{Mean}(x^2) \neq \text{Mean}(x)^2$$

3.

$$a) \quad f(x) = e^{-x}$$



$$F(x) = 1 - e^{-x}$$

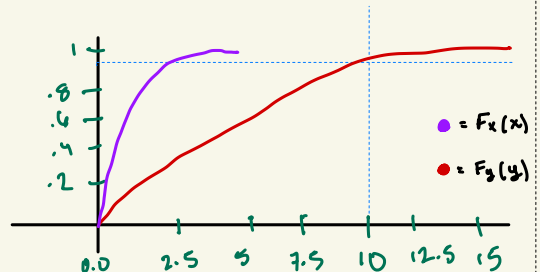


$$b) \quad \begin{aligned} &P(x \geq 3) + P(x < 3) = 1 \\ &\quad \quad \quad \hookrightarrow F(3) \rightarrow P(x \geq 3) = 1 - (P(x \leq 3)) = 1 - F(3) \\ &P(X \geq 3) = e^{-3} = 0.0498 \quad \quad \quad \therefore 1 - (1 - e^{-3}) \\ &P(x \leq 35) = 1 - e^{-35} = 1 \quad \quad \quad e^{-3} \end{aligned}$$

$$c) \quad Y = 1 + 3X \quad a=1; \quad b=3$$

$$\begin{aligned} F_Y(y) &= F_X\left(\frac{y-1}{3}\right) \\ &= 1 - e^{-\left(\frac{y-1}{3}\right)} \end{aligned}$$

$$\begin{aligned} P(Y < 10) &= F_Y(10) = 1 - e^{-\left(\frac{10-1}{3}\right)} \\ &= 1 - e^{-3} = .9502 \end{aligned}$$



4. a) $F(x) = \frac{1}{1+e^{-x}}$

$$f(x) = \frac{1}{1+e^{-x}} \cdot \left(1 - \frac{1}{1+e^{-x}}\right)$$

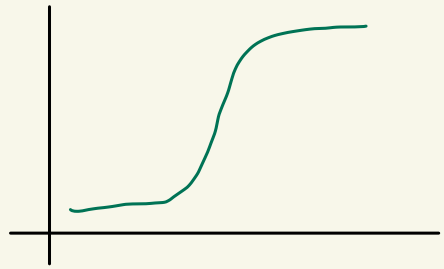
$$= \frac{1}{1+e^{-x}} \cdot \left(\frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}}\right)$$

$$= \frac{1}{1+e^{-x}} - \left(\frac{e^{-x}}{1+e^{-x}}\right) = \frac{e^{-x}}{(1+e^{-x})^2}$$

$f(x)$



$F(x)$



b)

$$F(0.8) = \frac{1}{1+e^{-.8}} = .69$$

$$P(.8) = 1 - .69 = .31$$

$$F(.3) = \frac{1}{1+e^{-.3}} = .5744$$

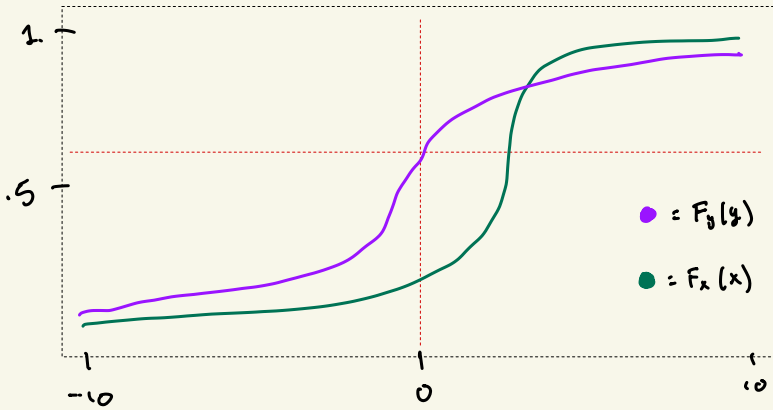
$$P(.3) = 1 - .5744$$

$$c) \quad Y = 2x - 1 \quad a = -1 \quad b = 2$$

$$F_Y(y) = F_X\left(\frac{y+1}{2}\right)$$

$$= \frac{1}{1 + e^{-\left(\frac{y+1}{2}\right)}}$$

$$P(Y < 0) = F_Y(0) = \frac{1}{1 + e^{-\left(\frac{0+1}{2}\right)}} = 0.6225$$



5)

$$a) \quad F(m) = 1 - e^m = \frac{1}{2}$$

$$\Rightarrow e^m = \frac{1}{2}$$

$$\ln \quad \ln$$

$$m = \ln(1/2) \approx -0.6931$$

$$b) \quad F(m) = \frac{1}{1 + e^{-m}} = \frac{1}{2}$$

$$\Rightarrow 1 + e^{-m} = 2$$

$$\ln \quad \ln$$

$$\left. \begin{array}{l} -m = \ln(1) \\ -m = 0 \\ m = 0 \end{array} \right\}$$

$$c) 1 - e^x - x = \mu$$

$$1 - \mu = e^x - x$$

ln ln

$$x = -\ln(1 - \mu)$$

$$F^{-1}(\mu) = -\ln(1 - \mu)$$

check 0.5

$$F^{-1}(0.5) = -\ln(0.5) \\ = \ln(2) \checkmark$$

$$d) \frac{1}{1 + e^{-x}} = \mu$$

$$1 = \mu(1 + e^{-x})$$

$$\frac{1}{\mu} = 1 + e^{-x}$$

$$\frac{1}{\mu} - 1 = e^{-x}$$

$$\ln \frac{1 - \mu}{\mu} = e^{-x}$$

ln ln

$$-x = \ln\left(\frac{1 - \mu}{\mu}\right)$$

$$x = -\ln\left(\frac{1 - \mu}{\mu}\right) = \ln\left(\frac{\mu}{1 - \mu}\right)$$

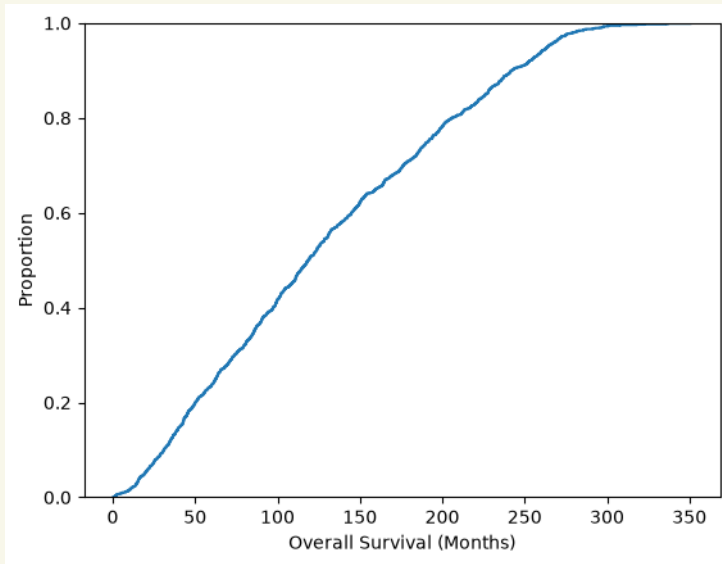
$$F^{-1}(\mu) = \ln\left(\frac{\mu}{1 - \mu}\right)$$

Check 0.5

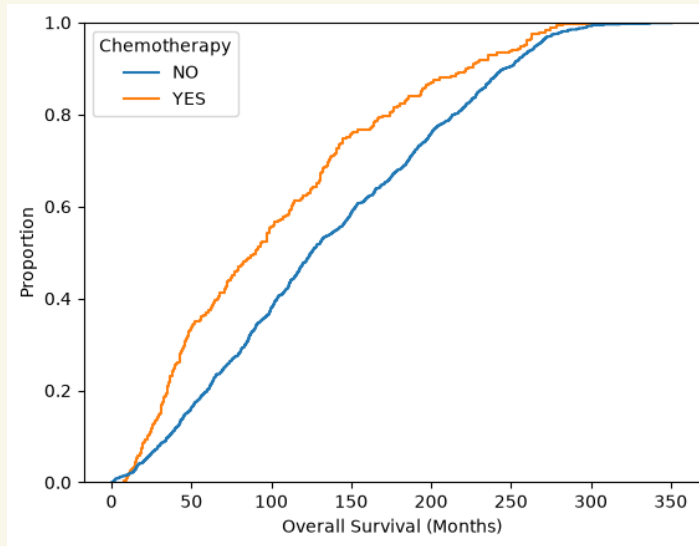
$$F^{-1}(0.5) = \ln\left(\frac{0.5}{0.5}\right) = \ln(1) = 0 \checkmark$$

b)

a)



b)



Chemotherapy
Curve sits on the
left of the no-chemo
curve. This means
chemo patients
reach the same
position @ shorter
time interval. So,
this plot predicts
shorter and not
longer survival w/
chemo.

c) The graph shows a shorter survival for the chemo group but from what we know about chemo it actually improves survival. The missing gap has a selection bias. Chemo goes to patients who typically have more advanced disease, so since they are more sick we can point the lower survival of chemo to the starting point and not the drug.